

# PAPER\_236: UQFF Learning Assessment Evolution\_B — Framework Meta-Assessment Advancement Score

**Author:** Daniel T. Murphy **Framework:** UQFF v4.9 (Star-Magic) **Session:** 59 (grok\_share\_8d951e12.txt second-pass — Doc 9) **Date:** March 2026 **Classification:** Novel — First Framework-Level Meta-Assessment Calculator in UQFF Pipeline **Status:** Proof-Quality Whitepaper **CP3 Class:** UQFFLearningAdvancementCalculator

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## Abstract

This paper documents the UQFF Learning Assessment Evolution\_B module, the ninth in a series of 500+ UQFF code files. Unlike all preceding modules (which target specific astrophysical objects), this is the first **framework-level meta-assessment calculator** in the CP1/CP2/CP3 pipeline. It computes a structured advancement score aggregating three orthogonal metrics: physical regime diversity, dynamical term novelty, and framework scalability. The advancement formula is:

$$\text{advancement} = \frac{\text{diversity\_score} + \text{dynamic\_score} + \text{scalability\_score}}{3.0} \times 100\%$$

The module aggregates parameter inputs from three consecutive UQFF examples: West-erlund 2 stellar wind, Pillars of Creation erosion, and Rings of Relativity lensing modulation — providing a comparative multi-system basis for evaluating framework maturity.

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## 1. Module Context

Source: UQFFLearningAssessment.h (Evolution\_B, October 08 2025 revision) Integration target: ziqn233h.cpp (base program, not included) Designed to instantiate as: UQFFLearningAssessment assess; double adv = assess.compute\_advancement();

This is the **first CP3 calculator not modelling an astrophysical object** — it models the UQFF framework's own learning progression, providing a quantitative measure of advancement per development session.

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## 2. Core Advancement Formula

$$\text{advancement} = \frac{d + D + s}{3.0} \times 100\%$$

where: -  $d$  = diversity\_score — number of distinct physical regimes demonstrated (default 3: stellar wind, erosion, lensing) -  $D$  = dynamic\_score — number of novel dynamic force/field terms introduced (default 3:  $a_{\text{wind}}$ ,  $E(t)$  erosion, lensing modulation) -  $s$  = scalability\_score — adaptability across spatial/temporal scales, normalised to [0, 1] (default 0.8)

### Default result:

$$\text{advancement} = \frac{3.0 + 3.0 + 0.8}{3.0} \times 100\% \approx 226.7\%$$

(Values  $> 100\%$  indicate multi-regime simultaneous coverage, i.e., framework is demonstrating super-linear progression.)

### 3. Parameters from Three Prior Examples

#### 3.1 Westerlund 2 (Stellar Wind Regime)

| Parameter                | Symbol                    | Default Value                             | Units             |
|--------------------------|---------------------------|-------------------------------------------|-------------------|
| Initial mass             | $M_{\text{wd2}}$          | $30,000 M_{\odot} = 5.967 \times 10^{34}$ | kg                |
| Radius                   | $r_{\text{wd2}}$          | (set per session)                         | m                 |
| Star formation timescale | $\tau_{\text{SF, wd2}}$   | $3.15 \times 10^{13}$                     | s                 |
| Wind density             | $\rho_{\text{wind, wd2}}$ | $1 \times 10^{-20}$                       | kg/m <sup>3</sup> |
| Wind velocity            | $v_{\text{wind, wd2}}$    | $2,000 \times 10^3$                       | m/s               |

#### 3.2 Pillars of Creation (Erosion Regime)

| Parameter                | Symbol                  | Default Value         | Units               |
|--------------------------|-------------------------|-----------------------|---------------------|
| Erosion factor (initial) | $E_0$                   | 0.3                   | —                   |
| Erosion timescale        | $\tau_{\text{erosion}}$ | $3.15 \times 10^{12}$ | s ( $\sim 0.1$ Myr) |

Erosion temporal model:  $E(t) = E_0 e^{-t/\tau_{\text{erosion}}}$

#### 3.3 Rings of Relativity (Gravitational Lensing Regime)

| Parameter                | Symbol              | Default Value          | Units           |
|--------------------------|---------------------|------------------------|-----------------|
| Einstein radius          | $r_{\text{rings}}$  | $1.54 \times 10^{22}$  | m               |
| Lensing amplitude factor | $L_{\text{factor}}$ | 1.2                    | —               |
| Hubble parameter at $z$  | $H_z$               | $2.18 \times 10^{-18}$ | s <sup>-1</sup> |

Lensing modulation:  $g_{\text{lens}} = U_{g1} \cdot H_z \cdot t + U_{g4}(1 + f_{\text{TRZ}}) \cdot L_{\text{factor}}$

### 4. Setter Architecture

The original C++ header provides per-variable setters:

```
void setDiversityScore(double v)           / addToVar / subtractFromVar -
same pattern for all params
```

This enables runtime update of any parameter without reconstructing the object, supporting iterative UQFF refinement workflows.

## 5. Novel Contribution

1. **First meta-assessment calculator** — evaluates UQFF progression itself, not an astrophysical object
  2. **Three-metric advancement formula** —  $\text{diversity} \times \text{dynamic} \times \text{scalability}$  normalised mean
  3. **Multi-example parameter aggregation** — single class absorbs parameters from 3 prior sessions
  4. **Advancement values > 100%** — framework intentionally designed to go beyond binary pass/fail
  5. **Supports quantitative session comparison** — advancement score can be tracked per git commit
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## 6. CP3 Implementation

**Class:** UQFFLearningAdvancementCalculator **Method:** `compute(dataset)`  $\rightarrow$  dict **Key output:** `advancement_pct` (float, %) **Available equations:** wind acceleration, erosion decay, lensing modulation, advancement formula

```
calc = UQFFLearningAdvancementCalculator()
result = calc.compute({'diversity_score': 3.0, 'dynamic_score': 3.0, 'scalability_score': 3.0})
# result['advancement_pct']  $\approx$  226.7 %
```

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## References

- Murphy, D.T. (2025). *UQFF Learning Assessment Evolution\_B Module*, UQFFLearningAssessment.h, October 08 2025
- `grok_share_8d951e12.txt`, Doc 9, lines 2993–3085 (UQFF Learning Assessment Evolution\_B)
- Westerlund 2: PAPER\_228 (Westerlund2MUGESTellarWindCalculator)
- Pillars of Creation: PAPER\_229 (PillarsOfCreationErosionMUGECalculator)
- Rings of Relativity: Prior session UQFFLensingModulationRingsCalculator